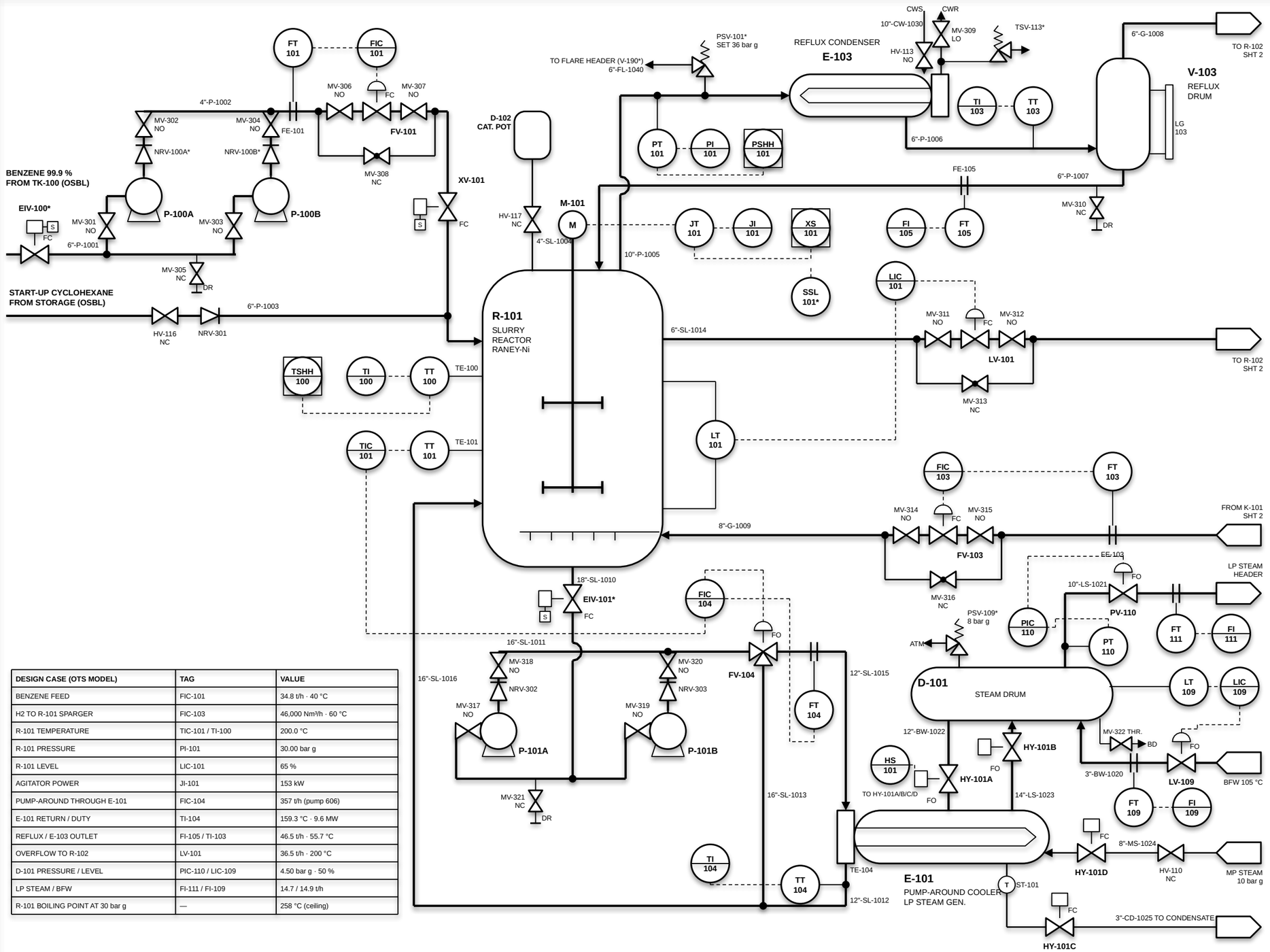


DESIGN CASE (OTS MODEL)	TAG	VALUE
BENZENE FEED	FIC-101	34.8 t/h · 40 °C
H2 TO R-101 SPARGER	FIC-103	46,000 Nm³/h · 60 °C
R-101 TEMPERATURE	TIC-101 / TI-100	200.0 °C
R-101 PRESSURE	PI-101	30.00 bar g
R-101 LEVEL	LIC-101	65 %
AGITATOR POWER	JI-101	153 kW
PUMP-AROUND THROUGH E-101	FIC-104	357 t/h (pump 606)
E-101 RETURN / DUTY	TI-104	159.3 °C · 9.6 MW
REFLUX / E-103 OUTLET	FI-105 / TI-103	46.5 t/h · 55.7 °C
OVERFLOW TO R-102	LV-101	36.5 t/h · 200 °C
D-101 PRESSURE / LEVEL	PIC-110 / LIC-109	4.50 bar g · 50 %
LP STEAM / BFW	FI-111 / FI-109	14.7 / 14.9 t/h
R-101 BOILING POINT AT 30 bar g	—	258 °C (ceiling)

LEGEND

- MAIN PROCESS LINE
- UTILITY / SECONDARY / BYPASS LINE
- INSTRUMENT PROCESS CONNECTION
- ELECTRONIC SIGNAL (4-20 mA / BUS)
- CROSSING WITHOUT CONNECTION (VERTICAL JUMPS) / TEE
- TT nnn — FIELD INSTRUMENT (TRANSMITTER, SWITCH)
- TIC nnn — DCS FUNCTION (INDICATION, CONTROL, ALARM)
- TSHH nnn — SIS FUNCTION (LOGIC SOLVER SIS-100)
- FC — CONTROL VALVE, DIAPHRAGM ACTUATOR; FAIL POSITION FC CLOSED / FO OPEN
- S / FC — ON/OFF VALVE WITH SOLENOID (SIS, EIV, HY)
- FO — THREE-WAY CONTROL VALVE
- MANUAL GATE VALVE (NO / NC / LO = NORMALLY OPEN / CLOSED / LOCKED OPEN)
- MANUAL GLOBE VALVE (BYPASS, THROTTLING)
- CHECK VALVE (FLOW LEFT TO RIGHT)
- SAFETY / THERMAL RELIEF VALVE
- FLOW ELEMENT (ORIFICE)
- DRAIN (DR) / VENT (V) VALVE WITH CAP
- T — STEAM TRAP
- OFF-PAGE CONNECTOR (SHEET REFERENCE BESIDE IT)

LINE NUMBER: SIZE"-SERVICE-NNNN (e.g. 8"-G-1009)
SERVICE: P PROCESS LIQUID / TWO-PHASE · G HYDROGEN GAS
SL SLURRY (Raney-Ni) · BW BOILER FEED WATER · LS LP STEAM
MS MP STEAM · CD CONDENSATE · BD BLOWDOWN · CW COOLING
WATER · FL FLARE · N NITROGEN. NNNN: 1nnn SHEET 1, 2nnn SHEET 2
TAGS: MV MANUAL VALVE · NRV CHECK VALVE · HV OPERATOR
HAND VALVE OF THE OTS · HY VALVE MOVED BY A DCS SELECTOR
* = RECOMMENDED ADDITION (SAFETY LAYERS TAB, SL-Rn);
NOT IN THE OTS MODEL: HOLD FOR APPROVAL

NOTES:
1 DESIGN CASE = OTS MODEL AT DESIGN, NOISE OFF: benzene 34.8 t/h (0.1238 kmol/s), R-101 200 °C / 30.0 bar g, 98 % conversion in R-101, E-101 9.6 MW raising 14.7 t/h LP steam at 4.5 bar g, boil-up condensed in E-103 and refluxed (46.5 t/h at 55.7 °C). Balances checked in eng_check.py before drawing: total 37.78 t/h in = out (+0.01 %), H2 and CH4 within 0.01 %, R-101 heat 25.2 MW generated vs 25.1 MW removed (−0.4 %), D-101 BFW = steam + 1 % blowdown.
2 LINE SIZES ARE PRELIMINARY, FROM VELOCITY (liquid suction ≤ 1–2 m/s, discharge ≤ 2–2.5 m/s, gas ≤ 12–15 m/s, steam ≤ 30 m/s; typical practice). TBC by hydraulic calculation; pipe classes not assigned (HOLD).
3 HYDROGEN LOOP DESIGN PRESSURE 36 bar g (assumed, Safety layers tab) = PSV-101 set; R-101 design temperature 250 °C (assumed). Relief loads not calculated (HOLD).
4 FIELD TRANSMITTERS (circle) are wired to the DCS (circle with bar) and, where drawn, separately to the SIS logic solver SIS-100 (circle in square). SIS outputs to XV / EIV solenoids and motor starters are not drawn: see the interlock schedule on sheet 2.
5 TE-100 / TT-100 (SIS) and TE-101 / TT-101 (BPCS) are separate elements in separate thermowells. TIC-101 is the master of FIC-104 (cascade). FV-104 is three-way: the rest of the P-101 flow bypasses E-101.
6 HS-101 (DCS selector) COOL: HY-101A/B open (E-101 shell on the D-101 thermosyphon), HY-101C/D shut. HEAT: HY-101A/B shut, HY-101C/D open, MP steam set by the operator on HV-110. A unit trip forces COOL (HY-101D shuts the MP steam: in the OTS model the trip closes HV-110). HY-101A-D and ST-101 are new tags of this drawing (HOLD: arrangement TBC).
7 ITEMS MARKED * are recommended additions from the Safety layers tab (SL-Rn), not in the OTS model: HOLD. Further SRS / LOPA recommendations not drawn: LSLL-101 on LT-101S, FSLL-103 on FT-103S, AT-100B, XV-105B, TSHH-107S on TE-107S, PIAH-101 (32.5 bar g on PT-101), TDA-101 (TT-100 vs TT-101).
8 MV-3nn and NRV-3nn are new numbers of this drawing (sheet 2: MV-4nn, NRV-4nn). Every pump has suction and discharge isolation and a discharge check valve. Pressure and level instruments c/w root, isolation and test valves (typ., not shown).
9 R-101 has no vent of its own: it is vented through E-103 / V-103 and the R-102 off-gas line to flare (HV-121, EDV-101, sheet 2). No isolation valve between R-101 and PSV-101.
10 Raney-Ni is pyrophoric: D-102 charged under liquid with an N2 blanket (CAT-1); HV-117 is opened only with D-102 full of liquid.
11 THE SCREEN P&ID OF THE OTS PAGE stays the reference for the DCS, the decks and the exercises. Where this drawing places an instrument differently it is listed as a difference (P&ID tab, alternative version, part 5), not as an error.

REV	DESCRIPTION	DATE
—	SCREEN P&ID IN OTS PAGE: CURRENT, USED IN DECKS	
0	ISSUED FOR TEACHING · ALT. VERSION, EPC STD	2026-10-26

CYCLOHEXANE REACTOR VER 1 — OTS TEACHING UNIT

P&ID — CYCLOHEXANE REACTOR VER 1 · ALTERNATIVE VERSION
SHEET 1: FEED · R-101 · E-101/D-101 · E-103/V-103

DRAWN	CHK'D	APP'D	DATE
Apinan Soottitantawat	—	—	2026-10-26

DWG NO.	REV	SCALE	SHEET
CHX-V1-PID-001	0	N.T.S.	1 OF 2